build-list

2026.08.23 - Strains to construct: 25 doubles + 20 triples

45 new strains. All are built from strains that already exist in the run-4 panel – no new single knockouts are needed, and **every one builds in parallel**: nothing waits on anything else.

The design on this sheet is `capped`. Six selection strategies were compared over the same 39 candidate triples; `capped` is the one being built, and every ID below comes from it. It takes the six best-predicted triples that involve a gene the model has no trigenic training data for, then fills the remaining fourteen from further down the ranking, which holds those genes to six of twenty. The comparison is under "Where this list came from"; the full argument is in [[experiments.W019-echo-crispr-array.next-strains-to-construct]].

Two conventions used throughout:

- **`*` on an ID** means the strain involves `YER079W` or `YLR312C-B`, the two genes with zero trigenic training records. Those predictions are extrapolation.
- **`routes`** is how many already-built doubles a triple can be made from. `1` means one parent only, so a failed cross has no backup until the matching new double is done.

Ordering suggestion: **T01–T15 first**, the fifteen at `routes` 1. T16–T20 each have a second route.

What already exists

From the run-4 order sheet, **12 singles, 13 doubles, 0 triples**:

```
experiments/W019-echo-crispr-array/data/run4_doubles_2026-08-06/
  Single-and-Double-KO-Strains-List-Order.csv
```

Singles s1–s12. Eleven are prediction nodes and all eleven are used below. `s11` YLR313C (SPH1) was built but was never a node in the inference panel, so it is in no predicted triple and is not on this plate.

id	ORF	common	id	ORF	common
s1	YJR060W	CBF1	s7	YLR104W	LCL2
s2	YPL081W	RPS9A	s8	YLL012W	YEH1
s3	YDR057W	YOS9	s9	YER079W	–
s4	YPL046C	ELC1	s10	YKL033W-A	–
s5	YBR203W	COS111	s11	YLR313C	SPH1 (not a panel node)
s6	YGL087C	MMS2	s12	YLR312C-B	–

Doubles d1–d13. Eight are parents for the triples below and must be re-measured on this plate; five are not used by any selected triple.

id	pair	used	serves triples (rank)
d1	YDR057W + YPL081W	yes	12, 13, 16
d2	YPL046C + YPL081W	no	–
d3	YER079W + YPL081W	no	–

id	pair	used	serves triples (rank)
d4	YLR312C-B + YPL081W	yes	1, 3, 4, 6
d5	YDR057W + YGL087C	yes	16, 39
d6	YDR057W + YER079W	no	–
d7	YDR057W + YLR312C-B	no	–
d8	YJR060W + YPL046C	yes	21, 31, 33
d9	YBR203W + YPL046C	yes	24, 28, 29, 31, 34
d10	YDR057W + YLL012W	yes	25, 38, 39
d11	YLL012W + YPL046C	yes	21, 24, 27
d12	YER079W + YJR060W	no	–
d13	YER079W + YLR312C-B	yes	2, 5

The 14th designed double, `YKL033W-A x YJR060W`, failed to construct and does not exist.

Part 1 – 25 double knockouts

Cross the two singles listed. Both parents already exist.

ID	gene 1	gene 2	ID	gene 1	gene 2
D01	YBR203W (COS111)	YDR057W (YOS9)	D14	YGL087C (MMS2)	YPL046C (ELC1)
D02	YBR203W (COS111)	YGL087C (MMS2)	D15	YGL087C (MMS2)	YPL081W (RPS9A)
D03	YBR203W (COS111)	YJR060W (CBF1)	D16	YJR060W (CBF1)	YLL012W (YEH1)
D04	YBR203W (COS111)	YKL033W-A	D17	YJR060W (CBF1)	YLR104W (LCL2)
D05	YBR203W (COS111)	YLL012W (YEH1)	D18	YKL033W-A	YLL012W (YEH1)
D06	YBR203W (COS111)	YLR104W (LCL2)	D19*	YKL033W-A	YLR312C-B
D07*	YBR203W (COS111)	YLR312C-B	D20	YKL033W-A	YPL046C (ELC1)
D08	YBR203W (COS111)	YPL081W (RPS9A)	D21	YKL033W-A	YPL081W (RPS9A)
D09	YDR057W (YOS9)	YKL033W-A	D22*	YLL012W (YEH1)	YLR312C-B
D10*	YER079W	YLL012W (YEH1)	D23*	YLR104W (LCL2)	YLR312C-B
D11*	YER079W	YLR104W (LCL2)	D24	YLR104W (LCL2)	YPL046C (ELC1)
D12	YGL087C (MMS2)	YLL012W (YEH1)	D25	YLR104W (LCL2)	YPL081W (RPS9A)
D13*	YGL087C (MMS2)	YLR312C-B			

Part 2 – 20 triple knockouts

`build from` is an **existing** double from run 4; cross it with the single in `x third single`. `routes` counts the already-built doubles this triple could be made from; start with the fifteen at `1`.

ID	gene 1	gene 2	gene 3	build from	x third single	routes
T01*	YKL033W-A	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YKL033W-A	1
T02*	YER079W	YLL012W (YEH1)	YLR312C-B	YER079W + YLR312C-B	YLL012W (YEH1)	1
T03*	YLR104W (LCL2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YLR104W (LCL2)	1
T04*	YBR203W (COS111)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YBR203W (COS111)	1
T05*	YER079W	YLR104W (LCL2)	YLR312C-B	YER079W + YLR312C-B	YLR104W (LCL2)	1
T06*	YGL087C (MMS2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YGL087C (MMS2)	1
T07	YBR203W (COS111)	YDR057W (YOS9)	YPL081W (RPS9A)	YDR057W + YPL081W	YBR203W (COS111)	1
T08	YDR057W (YOS9)	YKL033W-A	YPL081W (RPS9A)	YDR057W + YPL081W	YKL033W-A	1
T09	YDR057W (YOS9)	YKL033W-A	YLL012W (YEH1)	YDR057W + YLL012W	YKL033W-A	1
T10	YGL087C (MMS2)	YLL012W (YEH1)	YPL046C (ELC1)	YLL012W + YPL046C	YGL087C (MMS2)	1
T11	YBR203W (COS111)	YLR104W (LCL2)	YPL046C (ELC1)	YBR203W + YPL046C	YLR104W (LCL2)	1
T12	YBR203W (COS111)	YKL033W-A	YPL046C (ELC1)	YBR203W + YPL046C	YKL033W-A	1
T13	YJR060W (CBF1)	YLR104W (LCL2)	YPL046C (ELC1)	YJR060W + YPL046C	YLR104W (LCL2)	1
T14	YBR203W (COS111)	YGL087C (MMS2)	YPL046C (ELC1)	YBR203W + YPL046C	YGL087C (MMS2)	1
T15	YBR203W (COS111)	YDR057W (YOS9)	YLL012W (YEH1)	YDR057W + YLL012W	YBR203W (COS111)	1
T16	YDR057W (YOS9)	YGL087C (MMS2)	YPL081W (RPS9A)	YDR057W + YGL087C	YPL081W (RPS9A)	2
T17	YJR060W (CBF1)	YLL012W (YEH1)	YPL046C (ELC1)	YJR060W + YPL046C	YLL012W (YEH1)	2
T18	YBR203W (COS111)	YLL012W (YEH1)	YPL046C (ELC1)	YBR203W + YPL046C	YLL012W (YEH1)	2
T19	YBR203W (COS111)	YJR060W (CBF1)	YPL046C (ELC1)	YBR203W + YPL046C	YJR060W (CBF1)	2
T20	YDR057W (YOS9)	YGL087C (MMS2)	YLL012W (YEH1)	YDR057W + YGL087C	YLL012W (YEH1)	2

* on an ID = involves a gene absent from our trigenic library

`YER079W` and `YLR312C-B` have **zero** trigenic records in the datasets the prediction model was trained on, so predictions involving them are extrapolation. Exposure is deliberately limited to **6 of 20 triples** (T01, T02, T03, T04, T05, T06) and **7 of 25 doubles** (D07, D10, D11, D13, D19, D22, D23).

If those two genes are later dropped, **14 triples survive and 17 of the 25 new doubles are still needed** (D25, `YLR104W + YPL081W`, would then feed no triple). Worth settling before starting the starred strains.

Where this list came from

There were 39 candidate triples. The figure shows which of them each of the six strategies would pick, strongest-predicted at the top. **Read the `capped` column: that is this sheet.** It takes the six best-predicted triples that involve a starred gene, then fills the rest from further down the list to hold the starred genes to six. `rank` would have put 17 of 20 on those genes and `balanced` 15; `capped` is the only design that holds the number to a chosen value while still covering all eleven genes.

How often each gene ends up in a triple, again reading the `capped` bars. It spreads across all eleven genes, range 2 to 8, rather than concentrating on a few.

Notes for the bench

- **Eleven singles are used** and all already exist: YBR203W (COS111), YDR057W (YOS9), YER079W, YGL087C (MMS2), YJR060W (CBF1), YKL033W-A, YLL012W (YEH1), YLR104W (LCL2), YLR312C-B, YPL046C (ELC1), YPL081W (RPS9A).

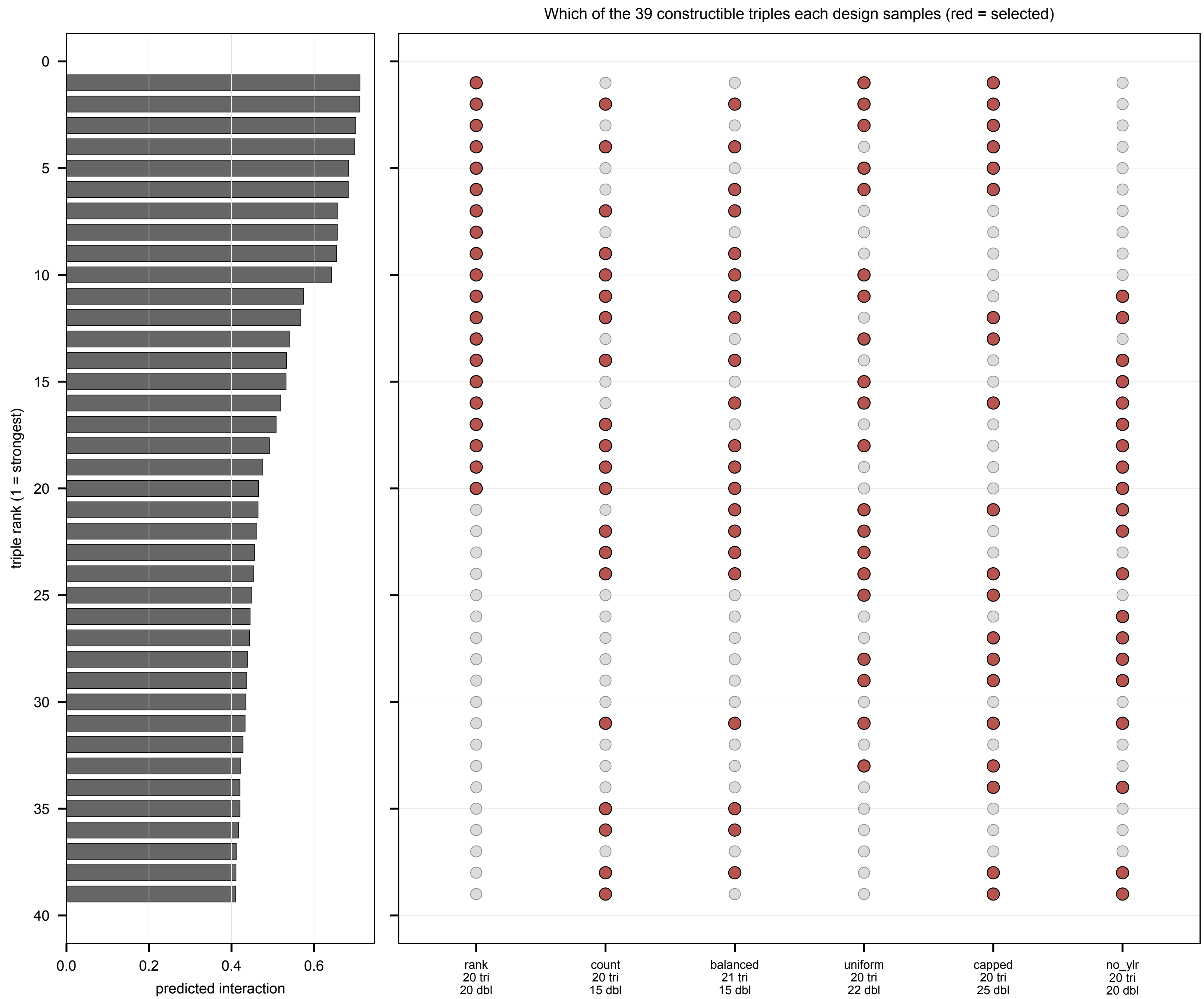

Figure 1: Which of the 39 candidate triples each design selects

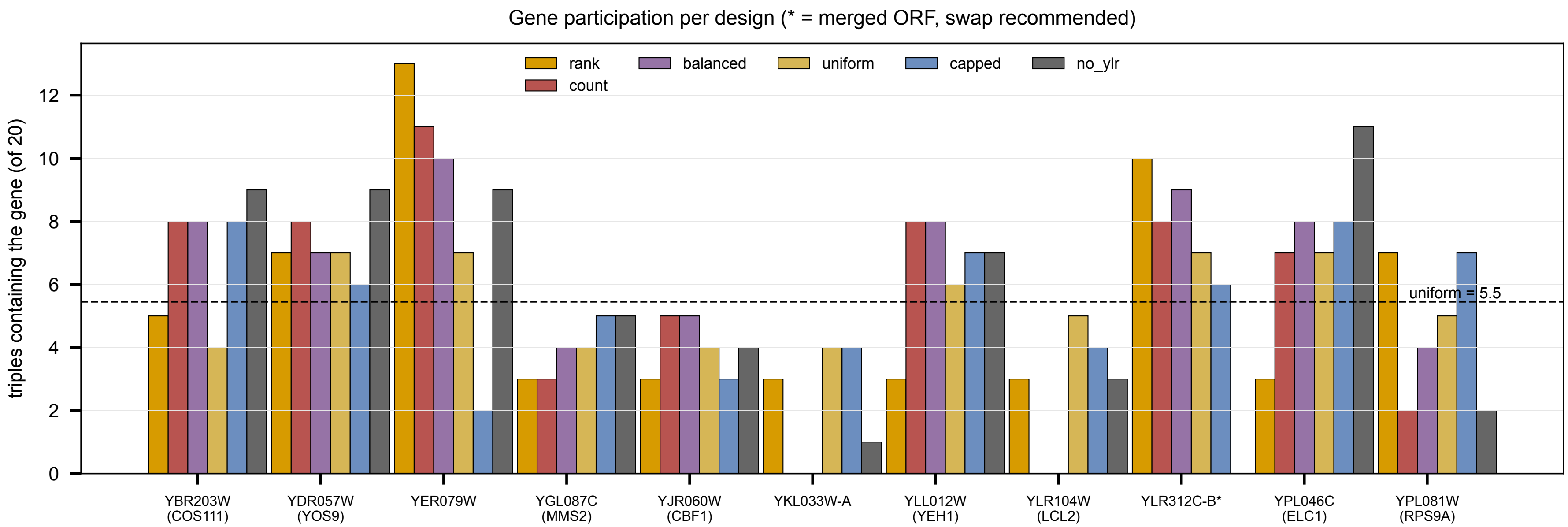

Figure 2: How often each gene appears

- **Do not attempt `YKL033W-A x YJR060W`.** It failed to construct previously and is deliberately not in this list. If anyone retries it, please record the attempt date and outcome – we have no record of when the original attempt happened.
- **`YLR104W` (LCL2) has no doubles in the current panel.** Six of the new doubles pair with it (D06, D11, D13, D17, D24, D25); those are the first coverage this gene has had.
- `YLR312C-B` is a merged/deprecated ORF; there is an open question about whether to use it or the adjacent real gene SPH1 (YLR313C). It appears in D07, D13, D19, D22, D23 and among the starred triples.
- The plating round also needs the existing singles plus **8 of the 13 existing doubles** as parents.

Plate

	count
singles	11
doubles	33 (8 existing + 25 new)
triples	20
WT (BY4741)	1
on plate	**65**
to construct	**45**

5 wells per strain on one 384 layout; 65 tubes at one pick per strain.

Why all 65 and not just the 20 triples. A trigenic interaction consumes seven measured fitnesses:

$$\tau_{abc} = f_{abc} - f_{ab}f_c - f_{ac}f_b - f_{bc}f_a + 2f_af_bf_c$$

so every triple needs its own fitness plus **all three** of its doubles and **all three** of its singles on the same plate. Verified against the selection: after the 45 strains are built, all 20 triples have all six supporting terms present, and the 65 strains listed here are exactly the closure of that requirement – no extra, none missing. What is bought is 20 computable tau values, not 20 fitness numbers.

Full rationale and measurement caveats: [[experiments.W019-echo-crispr-array.next-strains-to-construct]]